Local Spatial–Spectral Information-Integrated Semisupervised Two-Stream Network for Hyperspectral Anomaly Detection

Xiaoyi Wang, Liguo Wang, and Qunming Wang

Abstract— **Hyperspectral images (HSIs) always contain abundant spectral and spatial information. Most of the existing deep learning-based hyperspectral anomaly detection methods consider spectral differences between the background and anomalies, and the local spatial information is usually ignored. To make complete use of the spatial–spectral information, this article proposed a local spatial–spectral information-integrated semisupervised two-stream network (LS^{3}T-Net) for hyperspectral anomaly detection. The two-stream network comprises an adaptive convolution and fully connected network and a variational autoencoder (VAE). The adaptive convolution and fully connected network is used to extract the local spatial features of patches, while VAE is trained to learn spectral information close to the background pixels. Furthermore, the detection maps from the two-stream network are incorporated through a process combining the benefits of spatial learning and spectral learning. This enhances the ability to separate the background and anomalies and suppress the false alarm. The experimental results for six real HSI datasets reveal that LS^{3}T-Net can produce more accurate detection results than seven popular benchmark methods.**

Index Terms— **Convolution, hyperspectral anomaly detection, local spatial–spectral information, semisupervised, two-stream, variational autoencoder (VAE).**

I. Introduction

HYPERSPECTRAL anomaly detection has gained a great deal of interest in research and applications. For example, it can be employed in environmental monitoring, search and rescue, military reconnaissance, and other fields [1], [2], [3]. In recent decades, many hyperspectral anomaly detection methods have emerged. As the earliest method, the Reed–Xiaoli (RX) method proposed by Reed and Yu [4] assumes that the background obeys the Gaussian distribution and detects anomalies by calculating the Mahalanobis distance between the tested pixels and the estimated background model. Based on the RX method, Molero *et al.* [5] utilized a sliding dual-window strategy to estimate a local background model, which is dubbed as the LRX method. There are also other enhanced RX methods, such as the kernel RX method [6], the cluster kernel RX method [7], and the weighted RX method [8]. Li and Du [9] proposed a collaborative representation-based detector (CRD), which utilized the sliding dual-window strategy to construct a background dictionary to collaboratively represent the center pixel of the dual window. They further assumed that the background pixels could be well reconstructed, and the pixels with larger reconstruction errors are anomalies. Xu *et al.* [10] combined low-rank and sparse representation theories and proposed a hyperspectral anomaly method, in which low-rank representation is used to model the background part and sparsity-inducing regularization term is used to enhance the local representation. Zhang *et al.* [11] proposed a low-rank and sparse matrix decomposition (LRSMD)-based Mahalanobis distance method for hyperspectral anomaly detection (LSMAD), which uses prior low-rank knowledge of the background to generate the background model and Mahalanobis distance differences to assess anomalies. Cheng and Wang [12] introduced graph and total variation regularizations into the LRSMD model and proposed the graph and total variation regularized low-rank representation (GTVLRR) method to maintain the local geometric structure and spatial relationship of HSIs. Li *et al.* [13] proposed a method for hyperspectral anomaly detection based on prior-based tensor approximation (PTA). In PTA, the truncated nuclear norm and total variation regularization are used to constrain the spectral and spatial dimensions, respectively.

Deep learning has been extensively employed in remote sensing image processing due to its ability to mine deep-level data features and conduct strong nonlinear mapping [14], [15], [16]. In terms of hyperspectral anomaly detection, Li *et al.* [17] proposed a transferred deep convolutional neural network, in which a reference dataset with labeled samples was used to train the network. Xie *et al.* [18] proposed an autoencoder (AE) and adversarial-learning-based semisupervised background estimation (SBEM) model. In SBEM, the density-based spatial clustering of applications with noise (DBSCAN) method was used to search for background training samples. Li *et al.* [19] proposed a sparse coding-inspired generative adversarial network (GAN) for hyperspectral anomaly detection, named sparseHAD. Similar to SBEM, DBSCAN is used for background sample construction. Unlike SBEM, however, SparseHAD comprises two encoders and one

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decoder and uses hidden space data rather than original space data to detect anomalies. Jiang *et al.* [20] proposed a weakly supervised spectral constraint GAN for hyperspectral anomaly detection. Wang *et al.* [21] designed a fully convolutional AE with skip connections network and used reconstruction error as detection means. Jiang *et al.* [22] combined the AE and Gaussian mixture model (GMM) and proposed an end-to-end local-invariant autoencoding density estimation model (E2E-LIADE) for hyperspectral anomaly detection. Similar to [22], Xie *et al.* [23] also used AE and GMM to create a joint optimized anomaly detection model, which introduced a spectral distribution-aware module as local-invariant constraint.

Most of the existing deep learning-based hyperspectral anomaly detection methods are based on AE, and the training samples are spectral vectors, which do not completely use the local spatial information of HSIs. Due to the effects of illumination and platform altitude, the acquired HSIs commonly capture the same objects with different spectra. The spectral variation phenomenon is especially obvious at the boundary between the background and anomalies. Thus, the boundaries of abnormal objects could not be completely preserved, which affects the detection accuracy.

Integrating local spatial and spectral information is effective in hyperspectral image (HSI) processing [24], [25], [26]. In terms of hyperspectral anomaly detection, there are several recent studies combining spatial–spectral information. For example, Li *et al.* [27] proposed an end-to-end trainable deep one-class classification network for hyperspectral anomaly detection, which incorporated band selection, spectral–spatial feature learning, and anomaly detection into a framework. Li *et al.* [28] used the statistical decision of the similarity and difference between the target pixel and the surrounding background pixel as spatial constraint and used the spectral difference as spectral constraint, proposing a hyperspectral anomaly detection method based on the spatial and spectral constraints. To address the problem that the isolation forest method is difficult to detect high-dimensional data and local anomalies, Song *et al.* [29] used the Gabor filter and entropy rate segmentation to extract the spatial and spectral features and then fused the spatial and spectral detection maps through a linear function.

In this article, to improve the detection performance and completely use the local spatial and spectral information, we proposed a local spatial–spectral information-integrated semisupervised two-stream network for hyperspectral anomaly detection, which was defined as "LS^{3}T-Net." The main contributions of LS^{3}T-Net are given as follows.

1) To fully mine the local spatial information, we designed an adaptive convolution and fully connected (FC) network for spatial learning in the two-stream network. The adaptive network could automatically generate convolution layers with different depths according to the input patch size. This enhances the performance of anomaly detection at the boundary between background and anomalies and avoids manually designing parameters of convolutions.
2) We utilized the variational AE (VAE) to obtain spectral information close to background. From the perspective of probability distribution, the VAE can fit the background distribution more reliably. To further separate the background and anomalies and suppress false alarm, the exponential constrained nonlinear function was used to combine the advantages of spatial and spectral learning.

The remainder of this article is organized into four sections. Section II provides the related work. The proposed LS^{3}T-Net is described in detail in Section III. Section IV provides the experimental results based on six real HSI datasets and the comparison between LS^{3}T-Net and seven benchmark methods. The proposed LS^{3}T-Net and experimental results are further discussed in Section V. Finally, this article is concluded in Section VI.

II. Related Work

We define a 3-D HSI as $\mathbb{Y} \in \mathbb{R}^{H\times W\times D}$, where H and W represent the row and column, respectively, and D represents the dimension. Furthermore, let the original 3-D HSI transform into a 2-D matrix $\mathbf{Y} = [\mathbf{y}_1, \mathbf{y}_2, \ldots, \mathbf{y}_i, \ldots, \mathbf{y}_l] \in \mathbf{R}^{D\times L}$, where $L = H \times W$ represents the total number of pixels and $\mathbf{y}_i$ is the ith pixel with dimensions of D.

A. Mahalanobis Distance

The Mahalanobis distance [30] is a distance measure, which could resolve the inconsistency and related problems of various dimensions for Euclidean distance. In this article, we used the square Mahalanobis distance with the calculation as follows:

$$M_{y_i} = (\mathbf{y}_i - \boldsymbol{\mu})^T \boldsymbol{\Gamma}^{-1} (\mathbf{y}_i - \boldsymbol{\mu}) \tag{1}$$

where $\boldsymbol{\mu}$ and $\boldsymbol{\Gamma}$ are the mean and covariance matrix of $\mathbf{Y}$, respectively.

B. Activate Function

The activation function, also known as nonlinear mapping function, could enhance the nonlinear mapping ability of neural networks. In this article, the leaky rectified linear unit (LeakyReLU) [31] and Sigmoid [32] functions were considered. The LeakyReLU function is defined as

$$\text{LeakyReLU}(x) = \begin{cases} x, & x \geq 0 \\ ax, & x < 0 \end{cases} \tag{2}$$

where a is a small constant to solve the zero gradient problem of negative values. The sigmoid function is

$$\text{sigmoid}(x) = \frac{1}{1 + e^{-x}}. \tag{3}$$

III. Proposed Method

A. Extraction of Training Samples for Background

Ideally, in a semisupervised anomaly detection network, the training samples do not contain any anomaly pixels and contain as many background pixels as possible. As there is no prior anomaly spectrum, some unsupervised clustering

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methods are usually used to coarsely screen out the potential background pixels. In [18] and [19], DBSCAN was used to search background training samples. The advantage of DBSCAN is that it is insensitive to noise and can handle clusters with any size and shape. However, it is a parametric clustering method, and two parameters need to be set manually: radius and threshold. In addition, DBSCAN is limited when the data distribution density is uneven. In this article, to avoid heavy parameter adjustment, the Mahalanobis distance is used to select background training samples. In the HSI scene, the anomalies are few and with significant spectral difference from the background. We consider that the pixels with larger M_{y_i} values are more likely to be anomalies, while those with smaller M_{y_i} values are close to the background. Thus, to obtain relatively pure background training samples $\mathbf{B} = [\mathbf{b}_1, \mathbf{b}_2, \ldots, \mathbf{b}_i, \ldots, \mathbf{b}_q] \in \mathbf{R}^{D\times Q}(Q < L)$, we selected the first 90%–95% pixels with the smallest M_{y_i} values. Subsequently, the local image patches are constructed with the candidate pixels as the center in the original HSI. For a patch $\mathbf{p}_i \in \mathbb{R}^{P\times P\times D}$ centered at pixel $\mathbf{b}_i$, where P is the patch size generally set to an odd number (5, 7, and 9). Notably, when the candidate pixel is located at the edge of the HSI, the image needs to be expanded to ensure the sufficiency of train samples.

B. Proposed LS^{3}T-Net

The proposed LS^{3}T-Net is a semisupervised two-stream network, which comprises an adaptive convolutional and FC network and a VAE. For convenience, we denoted these as upstream and downstream networks.

1) Upstream Network: We assumed that the spatial adjacent pixels in HSIs have similar spectral characteristics. As the anomalies are few and with obvious spectral differences from the background, the background could be well weighted or collaborative representation by adjacent pixels; however, anomalies could not. In addition, there are obvious spectral variations at the boundary between background and anomalies. In general, the center of the abnormal object has a stronger spectral difference from the background, while the boundary shows a weaker spectral difference. Therefore, the anomalies at the boundary cannot be detected easily, which affects the overall accuracy. To solve this problem, we design an adaptive convolutional and FC network for local spatial feature learning to preserve boundary pixels. The convolution layers are used to extract the local spatial information of patches, and the FC layers integrate the extracted features into spectral vectors. In particular, to make complete use of all the spatial features of the original image patch, we use convolution layers to gradually extract the deep features of patches, and the padding and stride of each convolution layer are set to 0 and 1, respectively. We assumed that in the train phase, the center pixel of a patch is the background pixel, and most of the adjacent pixels are background pixels. Thus, after applying the convolution layers, the new pixel value is also similar to the background.

The adaptive network refers to determining the number of convolution layers according to the input patch size. The patch size is an odd number, it is expected that the spatial size of feature image after each convolution layer will be two pixels less than that of the previous layer until it is reduced to 1×1. For example, when the input patch size is 5×5, two convolution layers are required to obtain feature images with spatial sizes of 3×3 and 1×1, and when the input image patch size is 7×7, three convolution layers are required to obtain feature images with spatial sizes of 5×5, 3×3, and 1×1. Thus, when the input patch size is P, there are $n = ((P-1)/2)$ convolutional layers. After applying each convolution layer, the batch normalization (BN) [33] layer is used to normalize data in batch, and the LeakyReLU function is applied as an activation function. Each FC layer contains several nodes, and the LeakyReLU function and dropout layer are applied after the FC layer. The fundamental goal of the dropout layer is to prevent overfitting, especially when there are fewer training samples and more parameters. As presented in Table I, the convolution kernel size is set to 3×3. The parameters ε and momentum in the BN layer are set to 10^{-3} and 10^{-1}, respectively, and the dropout ratio is 0.5.

The upstream network will output a spectral vector $\hat{\mathbf{b}}_i^1 \in \mathbf{R}^{D\times 1}$. We expect that it is close enough to the center pixel $\mathbf{b}_i$. To minimize the reconstruction error between $\mathbf{b}_i$ and $\hat{\mathbf{b}}_i^1$, we design a reconstruction loss function for the upstream network. The loss function of the upstream network consists of two components: mean squared error (MSE) and smooth $\ell_1 - norm$. The concrete calculating methods are given as follows:

$$\mathrm{MSE}\left(\hat{\mathbf{b}}_i^1, \mathbf{b}_i\right) = \left\|\mathbf{b}_i - \hat{\mathbf{b}}_i^1\right\|_2 \tag{4}$$

$$\mathrm{Smooth}l_1\left(\hat{\mathbf{b}}_i^1, \mathbf{b}_i\right) = \begin{cases} \dfrac{1}{2}\left\|\mathbf{b}_i - \hat{\mathbf{b}}_i^1\right\|_2, & \text{if } \left|\mathbf{b}_i - \hat{\mathbf{b}}_i^1\right| < 1 \\ \left|\mathbf{b}_i - \hat{\mathbf{b}}_i^1\right| - \dfrac{1}{2}, & \text{otherwise.} \end{cases} \tag{5}$$

The reconstruction loss of the upstream network is expressed as

$$\mathrm{loss}_{\mathrm{up}} = \mathrm{MSE}\left(\hat{\mathbf{b}}_i^1, \mathbf{b}_i\right) + \mathrm{smooth}l_1\left(\hat{\mathbf{b}}_i^1, \mathbf{b}_i\right). \tag{6}$$

2) Downstream Network: The goal of the downstream network is to obtain the spectral distribution of the background samples. Inspired by GAN [34], [35], [36], for hyperspectral anomaly, most of the existing deep learning-based methods use AE as a generator and introduce a discriminator to improve the ability to reconstruct the background pixels [18], [19]. However, the independent AE could not learn the real data distribution from the perspective of generation. VAE is a directed probabilistic graphical model whose posterior is approximated by a neural network, thereby forming an AE-like architecture [37], [38]. The difference is that AE conducts single value mapping, while VAE could realize distribution mapping, which could better represent the data distribution. The VAE model has shown great potential in generating a variety of complex data, including handwritten digital image [39], face image [40], and prediction from static image [41]. In this article, we use the VAE as spectral learning network.

The architecture features of the downstream network are listed in Table I; notably, the last layer of the decoder uses the Sigmoid function. For each input potential background

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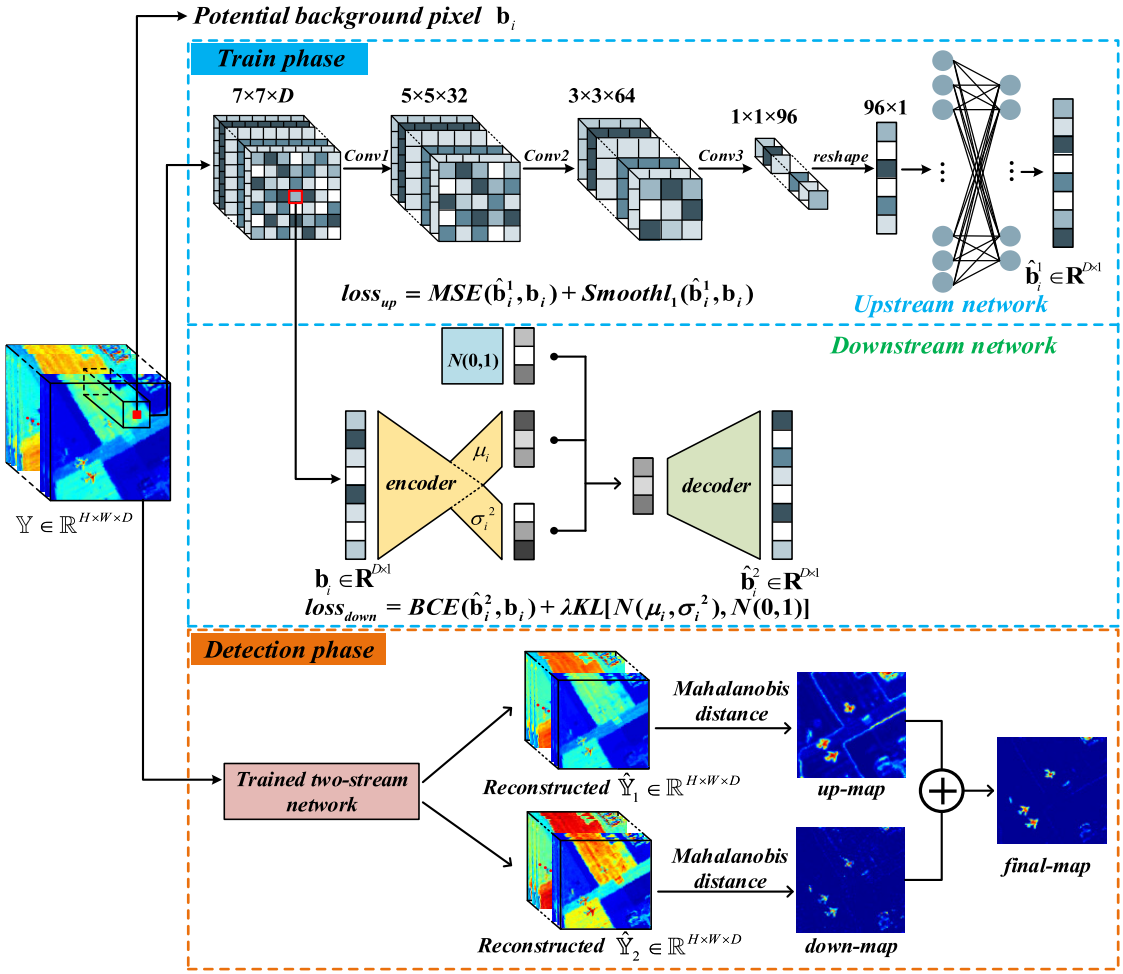


Fig. 1. Flowchart of the proposed LS^{3}T-Net method.

TABLE I

ARCHITECTURE FEATURES OF THE PROPOSED LS^{3}T-NET MODEL

Upstream network				
Layers	Type	Kernel shape	Input and output dimension	Padding
CONV1	Conv2D+BatchNorm2d+LeakyReLU	3×3	(D, 32)	0
CONV2	Conv2D+BatchNorm2d+LeakyReLU	3×3	(32, 64)	0
…	…	…	…	…
CONVn	Conv2D+BatchNorm2d+LeakyReLU	3×3	(32(n-1), 32n)	0
FC1	Linear+LeakyReLU+dropout(0.5)	-	(32n, K_1)	-
FC2	Linear+LeakyReLU+dropout(0.5)	-	(K_1, D)	-
Downstream network				
Layers	Type	Kernel shape	Input and output dimension	Padding
FC1	Linear+LeakyReLU	-	(D, 128)	-
FC2	Linear+LeakyReLU	-	(128, 64)	-
FC3	Linear+LeakyReLU	-	(64, 32)	-
$FC4_1$	Linear	-	(32, K_2)	-
$FC4_2$	Linear	-	(32, K_2)	-
FC5	Linear+LeakyReLU	-	(K_2, 32)	-
FC6	Linear+LeakyReLU	-	(32, 64)	-
FC7	Linear+LeakyReLU	-	(64, 128)	-
FC8	Linear+Sigmoid	-	(128, D)	-

pixel $\mathbf{b}_i$, the encoder of the VAE generates its corresponding mean μ_i and variance σ_i^2 to determine a normal distribution $N(\mu_i, \sigma_i^2)$. The hidden layer code $\mathbf{z}_i$ is expressed as

$$\mathbf{z}_i = \mu_i + \varphi \times \sigma_i \tag{7}$$

where φ is a random sample from the normal distribution $N(0, 1)$. Finally, a spectral vector $\hat{\mathbf{b}}_i^2 \in \mathbf{R}^{D\times 1}$ is reconstructed by the decoder of VAE, and $\hat{\mathbf{b}}_i^2$ is calculated as

$$\hat{\mathbf{b}}_i^2 = f(\boldsymbol{W}\mathbf{z} + \delta) \tag{8}$$

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TABLE II
SOME IMAGE CHARACTERISTICS OF THE SIX DATASETS

Datasets	Location	Sensor	Spatial size	No. of Bands
Dataset I	Sandiego	AVIRIS	100×100	189
Dataset II	Los angeles	AVIRIS	100×100	205
Dataset III	Gulfport	AVIRIS	100×100	191
Dataset IV	Texas Coast	AVIRIS	100×100	204
Dataset V	Sandiego	AVIRIS	100×100	189
Dataset VI	California	HYDICE	80×120	175

where $\boldsymbol{W}$ and δ denote the weight matrix and the bias term of the decoder, respectively, and $f(\bullet)$ is the decoding process. We expect that not only $\mathbf{b}_i$ and $\hat{\mathbf{b}}_i^2$ are close but also $N(\mu_i, \sigma_i^2)$ is close to the standard normal distribution $N(0, 1)$. In this article, the binary cross entropy (BCE) is used as the reconstruction loss of VAE. Also, the Kullback–Leibler (KL) divergence is introduced to measure the distribution difference. The calculation formula of BCE and KL divergence is given as follows:

$$\mathrm{BCE}\left(\hat{\mathbf{b}}_i^2, \mathbf{b}_i\right) = \mathbf{b}_i \log\left(\hat{\mathbf{b}}_i^2\right) + (1 - \mathbf{b}_i)\log\left(1 - \hat{\mathbf{b}}_i^2\right) \quad (9)$$

$$\mathrm{KL}\left(N\left(\mu_i, \sigma_i^2\right), N(0, 1)\right) = \frac{1}{2}\left(-\log \sigma_i^2 + \mu_i + \sigma_i^2 - 1\right). \quad (10)$$

The total loss function of VAE is

$$\mathrm{loss}_{\mathrm{down}} = \mathrm{BCE}\left(\hat{\mathbf{b}}_i^2, \mathbf{b}_i\right) + \lambda \mathrm{KL}\left(N\left(\mu_i, \sigma_i^2\right), N(0, 1)\right) \quad (11)$$

where λ is a tradeoff parameter set as 10^{-3}.

3) Anomaly Detection: The original HSI $\mathbb{Y}$ is inputted into the trained two-stream model, and two reconstructed HSIs $\hat{\mathbb{Y}}_1$ and $\hat{\mathbb{Y}}_2 \in \mathbb{R}^{M \times N \times D}$ are output from upstream and downstream networks, respectively. Calculating the Mahalanobis distance for $\hat{\mathbb{Y}}_1$ and $\hat{\mathbb{Y}}_2$, two initial detection maps [$\mathbf{I}_1$ (up-map) and $\mathbf{I}_2$ (down-map)] can be obtained. The calculation is consistent with (1).

To fully combine the advantages of spatial learning and spectral learning, we use the exponential constrained nonlinear function to fuse the two initial detection maps. Before fusion, the two initial detection maps were normalized by the max–min normalization method

$$\mathbf{I}_{\mathrm{up}} = \frac{\mathbf{I}_1 - \min(\mathbf{I}_1)}{\max(\mathbf{I}_1) - \min(\mathbf{I}_1)}$$
$$\mathbf{I}_{\mathrm{down}} = \frac{\mathbf{I}_2 - \min(\mathbf{I}_2)}{\max(\mathbf{I}_2) - \min(\mathbf{I}_2)} \quad (12)$$

where $\min(\bullet)$ and $\max(\bullet)$ represent the minimum and maximum values of the data, respectively. The final detection map (final-map) $\mathbf{I}_{\mathrm{final}}$ can be calculated by

$$\mathbf{I}_{\mathrm{final}} = \left(1 - \mathrm{e}^{-k\mathbf{I}_{\mathrm{down}}}\right)\mathbf{I}_{\mathrm{up}} \quad (13)$$

where k is a tradeoff parameter set to 5. The influence of k value on the detection map will be further analyzed in Section III-C. The flowchart of the proposed semisupervised two-stream model is shown in Fig. 1, considering $P = 7$ as an example. In Table I, K_1 and K_2 represent the number of the hidden nodes of upstream and downstream networks, respectively. CONVi and FCi ($i \in N_+$) are the ith convolution and FC layer, respectively.

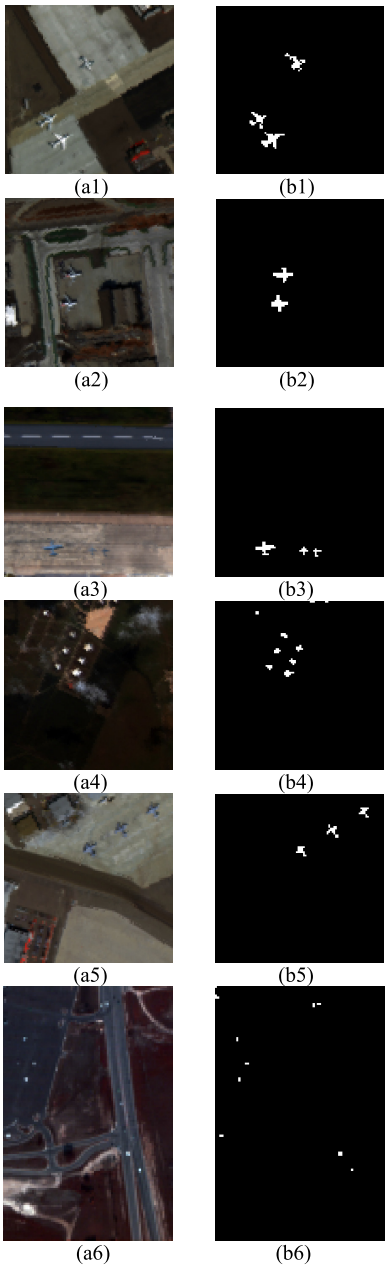


Fig. 2. Six datasets used in the experiments. (a1)–(a6) False-color images (bands 37, 18, and 8 as RGB) of the six real HSI datasets and (b1)–(b6) corresponding ground-reference maps of anomalies.

IV. EXPERIMENTS

A. HSI Datasets and Experimental Setup

Six real HSI datasets were applied in the experiments. The most important image characteristics of the six datasets are presented in Table II. The acronyms AVIRIS and HYDICE represent Airborne Visible/Infrared Imaging Spectrometer and Hyperspectral Digital Imagery Collection Experiment, respectively. Fig. 2 shows the false-color images and ground-reference maps of the six datasets.

For comparison, seven popular hyperspectral anomaly detection methods were selected as the benchmark methods. These include RX [4], CRD [9], LSMAD [11], GTVLRR [12], PTA [13], SparseHAD [19], and E2E-LIADE [22]. In the experiments, the dual-window sizes of CRD were set to (21,

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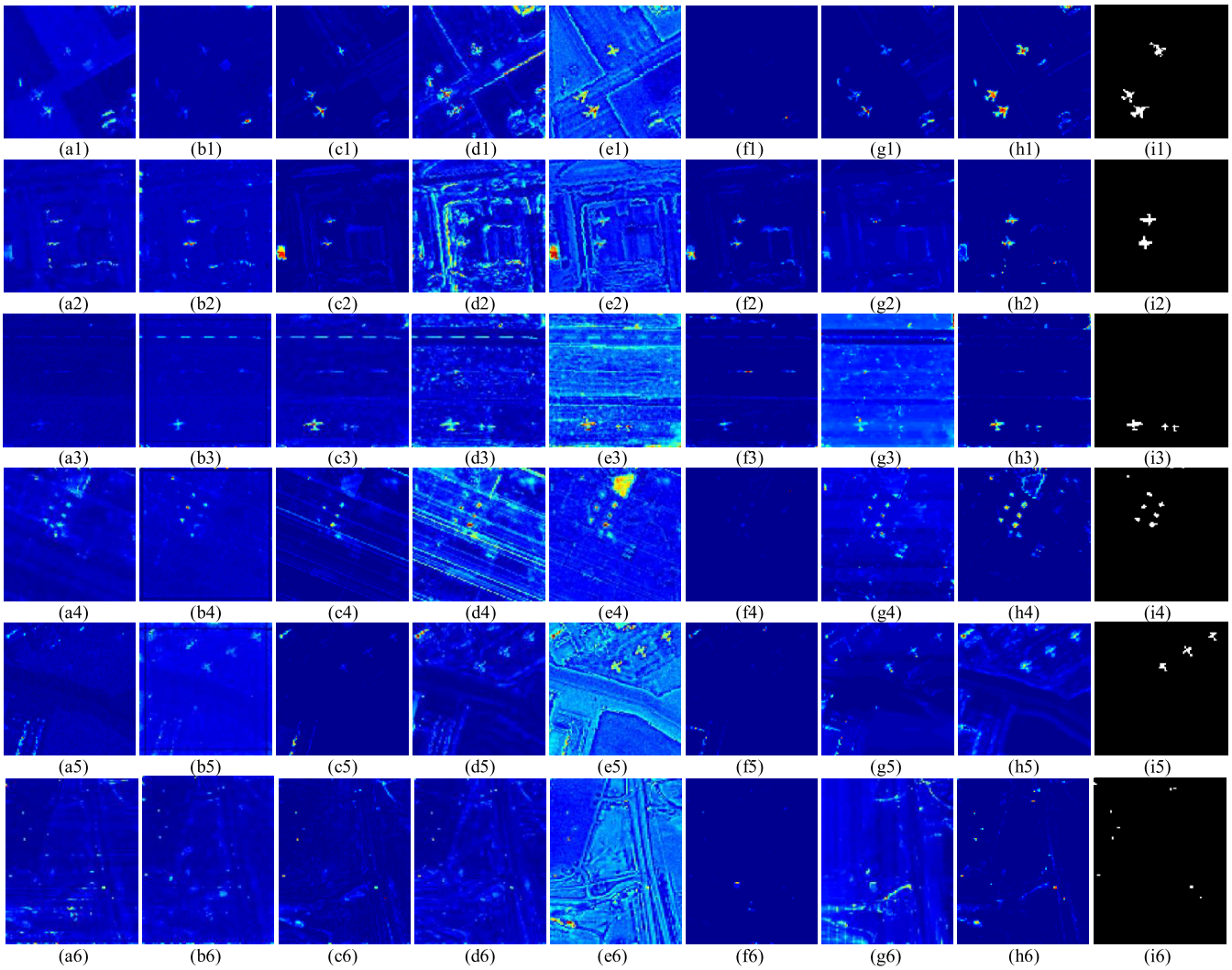


Fig. 3. Detection maps based on the eight methods for the six HSI datasets. (a)–(g) (Left to right) Detection maps for RX, CRD, LSMAD, GTVLRR, PTA, SparseHAD, E2E-LIADE and LS^{3}T-Net, respectively. (i) Corresponding ground-reference map. Lines 1–6 represent datasets I–VI, respectively.

TABLE III

AUC VALUES (P_f, P_d)/(τ, P_f) AND RUNNING TIME (IN SECONDS) OF THE EIGHT METHODS FOR THE SIX DATASETS

	RX	CRD	LMSAD	GTVLRR	PTA	SparseHAD	E2E-LIADE	LS^{3}T-Net
Dataset I	0.9403/0.0588	0.9664/0.0267	0.9710/0.0160	0.9311/0.0781	0.9442/0.1810	0.9816/**0.0054**	0.9846/0.0124	**0.9940**/0.0117
	1.5587	11.3878	8.9300	181.9313	29.2450	472.6689	4621.9743	547.3022
Dataset II	0.8404/0.0516	0.9758/0.0578	0.9561/0.0263	0.8701/0.1396	0.9128/0.1544	0.9572/0.0170	0.8703/0.0373	**0.9904**/**0.0092**
	0.5456	28.7961	10.2379	206.5908	37.6736	456.3589	8184.7821	404.3541
Dataset III	0.9526/0.0246	0.9767/0.0499	0.9899/0.0493	0.9865/0.0878	0.9717/0.2134	0.9746/0.0140	0.9784/0.1568	**0.9949**/**0.0099**
	0.5605	9.7923	9.4180	210.1989	36.7206	459.7224	4000.0468	558.0518
Dataset IV	0.9907/0.0554	0.9947/0.0562	0.9693/0.0318	0.9537/0.1579	0.9806/0.1347	0.9923/**0.0055**	0.9786/0.0695	**0.9966**/0.0143
	0.9273	30.3984	10.4756	241.1211	43.4174	482.7026	5208.3327	475.5001
Dataset V	0.8885/0.0379	0.9865/0.0934	0.9905/0.0087	0.9800/0.0485	0.9874/0.2304	0.9736/**0.0075**	0.9910/0.0375	**0.9949**/0.0449
	0.9751	33.3282	8.8270	191.7638	46.3232	451.8619	4248.4059	833.2634
Dataset VI	0.9839/0.0503	**0.9937**/0.0477	0.9902/0.0267	0.9878/0.0518	0.8432/0.2139	0.9870/**0.0057**	0.7888/0.0893	0.9904/0.0117
	1.4300	8.0079	7.4603	220.5801	43.8612	448.1033	4526.4665	482.9954

19), (17, 15), (19, 17), (13, 5), (21, 17), and (17, 15) for datasets I–VI, respectively, and the tradeoff parameter was set to 10^{-6}. In LSMAD, the rank values of the six datasets were set to 2, 7, 20, 17, 3, and 8. As indicated in [12], the tradeoff parameters of GTVLRR are $\lambda = 0.05$, $\beta = 0.2$, and $\gamma = 0.02$. The tradeoff parameters of PTA are $\alpha = 1.7$, $\beta = 0.05$, and $\mu = 0.01$. Moreover, the truncate rank values were set to 17, 30, 30, 40, 40, and 40 for datasets I–VI, respectively. The learning rate and the number of hidden nodes in SparseHAD were set to 10^{-4} and 20, respectively. The tradeoff parameters of E2E-LIADE are $\lambda_1 = 3$, $\lambda_2 = 0.1$, and $\lambda_3 = 10^{-4}$. In the proposed LS^{3}T-Net, the patch size P and the number of two

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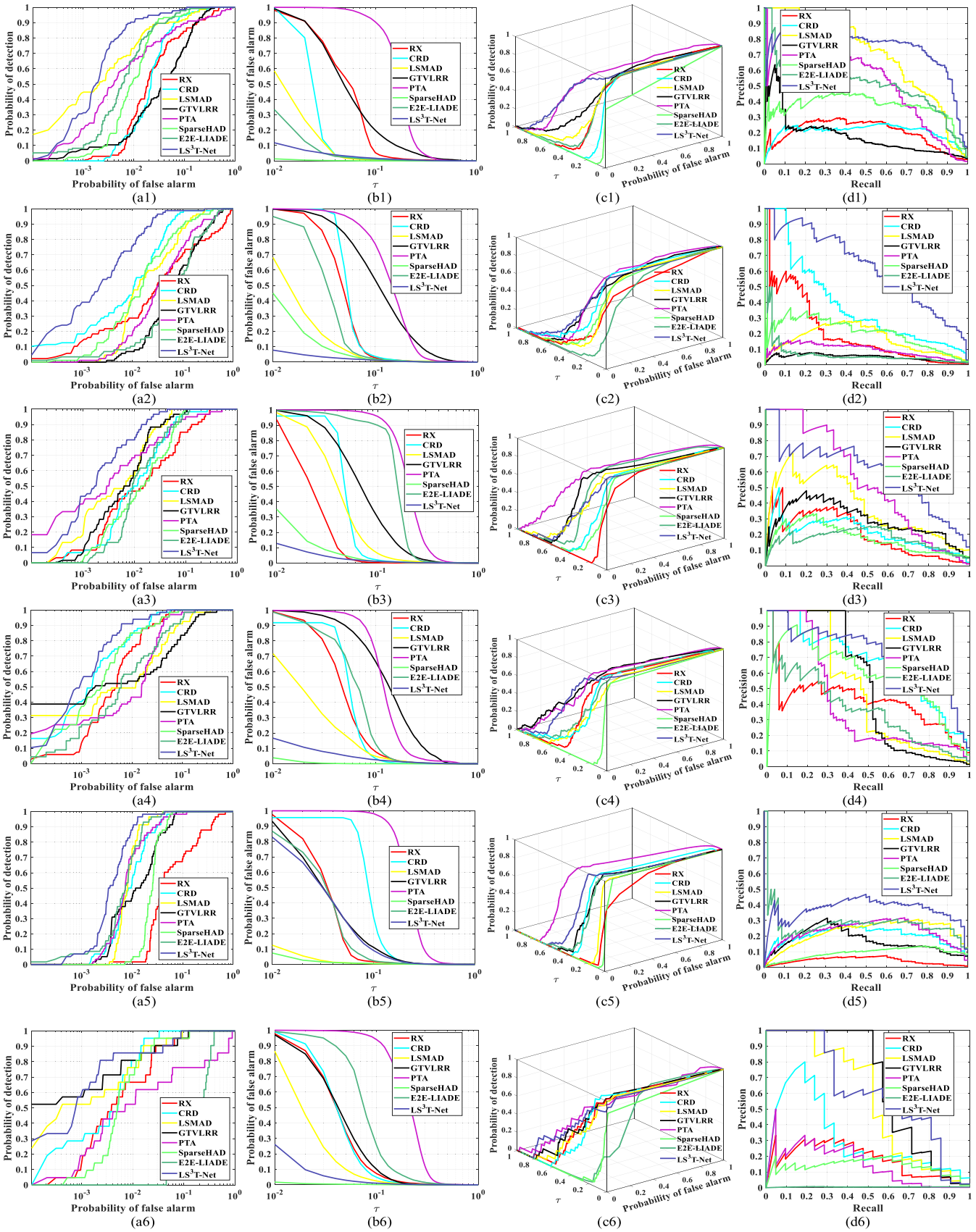


Fig. 4. ROC curves of the eight methods for the six real datasets. (a) 2-D ROC curves of (P_f, P_d), (b) 2-D ROC curves of (τ, P_f), (c) 3-D ROC curves, and (d) PR curves. Lines 1–6 are results for datasets I–VI, respectively.

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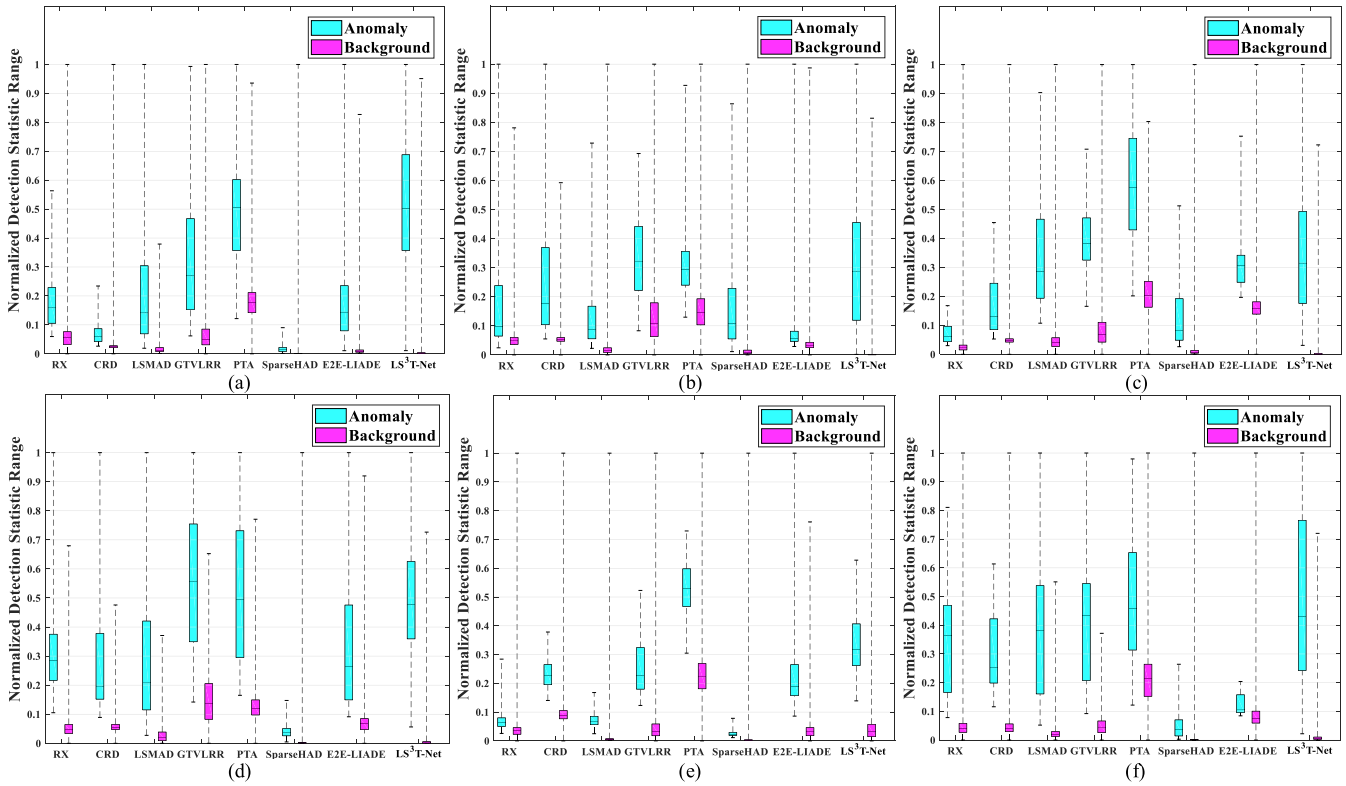


Fig. 5. Box-whisker plots for the eight anomaly detection methods for the six datasets. (a) Dataset I. (b) Dataset II. (c) Dataset III. (d) Dataset IV. (e) Dataset V. (f) Dataset VI.

hidden nodes K_1 and K_2 for the six datasets were set to (5, 1024, 4), (5, 1024, 4), (7, 1024, 2), (5, 2048, 4), (9, 2048, 2), and (7, 2048, 2). The learning rate was 10^{-4} for the two-stream network.

We use the area under the curve (AUC) values [42], the three-dimensional and two-dimensional receiver operating characteristic (ROC) curves [43], [44] and the precision–recall (PR) curve [45] as indices for accuracy evaluation. There are two versions of the two-dimensional ROC curves, ROC (P_f, P_d) and ROC (τ, P_f), where P_f and P_d are the probabilities of false alarm and detection and τ is the threshold. The AUC values indicate the area under the 2-D ROC curves, and the ideal values of AUC (P_f, P_d) and AUC (τ, P_f) are 1 and 0, respectively. In addition, the box-whisker plot is applied to investigate the separability between the background and anomalies.

The rest of this section consists of four parts. Sections III-B presents the anomaly detection results based on the eight methods for the six datasets. Section III-C analyzes the initial detection results of the proposed LS^{3}T-Net. Section III-D analyzes the influences of hyperparameters in LS^{3}T-Net on the detection results. Section III-E analyzes the influence of the proportions of training samples on experimental results.

B. Detection Results of the Six Datasets

Fig. 3 shows the detection maps based on the eight anomaly detection methods for the six datasets. LS^{3}T-Net shows great performance in both detecting anomalies and suppressing the background. Considering dataset I as an example, both CRD and SparseHAD fail to detect the position of abnormal aircraft. Although RX, LSMAD, GTVLRR, PTA, and E2E-LIADE could detect the position of abnormal aircraft, they could not adequately suppress the background, resulting in a greater false alarm rate, especially for GTVLRR and PTA. In addition, LS^{3}T-Net could effectively preserve the shape of anomalies. Another example is dataset II, and compared with other datasets, dataset II has more complex background distribution. These seven benchmark methods basically fail to preserve the shape of the two abnormal aircraft. However, as shown in Fig. 3(g2), LS^{3}T-Net could not only preserve the relatively complete shapes of the abnormal aircraft but also adequately reduce the impact of the complex background on the detection results. Although SparseHAD could also suppress background, it was unable to adequately differentiate the anomalies from the background, resulting in that there are no discernible brightness differences on the detection map.

Fig. 4 shows the three-dimensional and 2-D ROC curves for the six datasets based on the eight methods. It is observed that the 2-D ROC curves (P_f, P_d) of LS^{3}T-Net are located in the top-left corner in Fig. 4(a1)–(a5). Meanwhile, LS^{3}T-Net has smaller false alarm rate. As shown in Fig. 4(b2) and (b3), the 2-D ROC curves (P_f, τ) of LS^{3}T-Net are located in the bottom-left corner. Through the 3-D ROC curves, we could intuitively observe that the proposed LS^{3}T-Net method is the most satisfactory method overall. Two conclusions can be observed through PR curves: 1) in Fig. 4(d1)–(d6), except for dataset VI, the PR curves of the remaining datasets based on LS^{3}T-Net are located in the top-right corner, which indicates that LS^{3}T-Net has higher accuracy, and 2) generally, the ROC and PR curves show similar trends. For example, in Fig. 4(a2), LS^{3}T-Net is better than CRD, and CRD is

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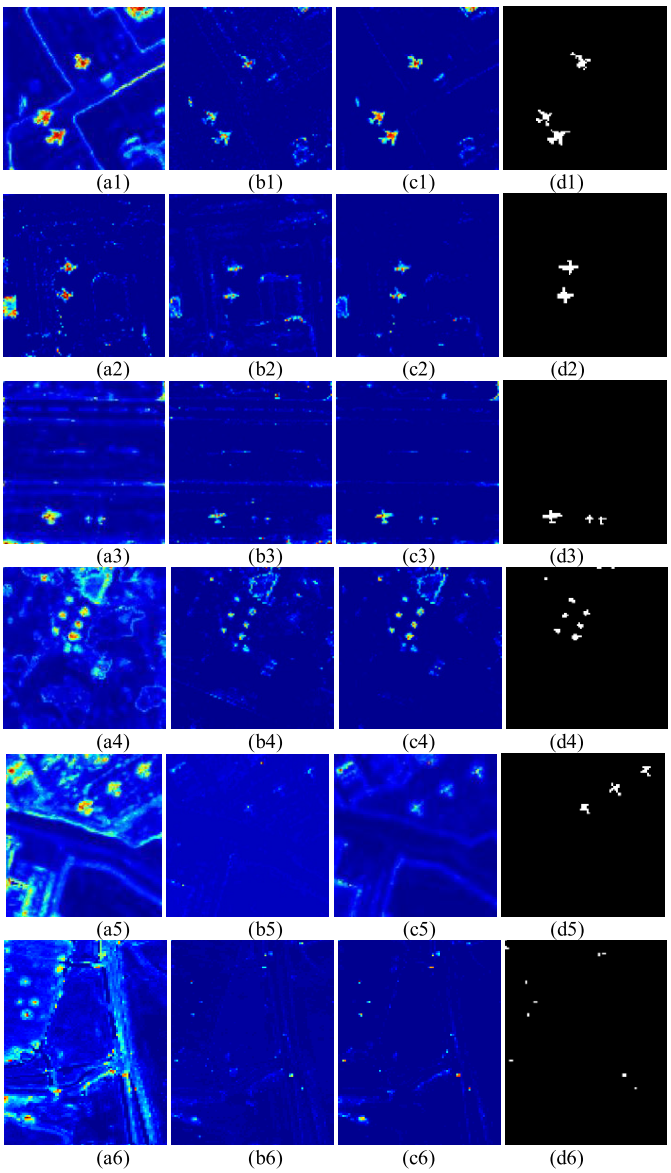


Fig. 6. Initial and final detection maps of LS^{3}T-Net for the six datasets. (a) $\mathbf{I}_{\text{up}}$, (b) $\mathbf{I}_{\text{down}}$, (c) $\mathbf{I}_{\text{final}}$, and (d) ground-reference maps. Lines 1–6 are detection maps for datasets I–IV, respectively.

better than SparseHAD. The order is the same as that shown in Fig. 4(d2). In addition, the box-whisker plot was used to further demonstrate the separability between the background and anomalies. In general, when there are a greater distance and less overlap between background and anomaly boxes, this method is considered to have greater separability between the background and anomalies. As shown in Fig. 5, the background boxes of LS^{3}T-Net close to 0, and there is a larger distance between the background boxes and the anomaly boxes for all six datasets.

The AUC values and running time based on the eight methods are listed in Table III, in which the most accurate results of AUC (P_f, P_d) and AUC (τ, P_f) for each dataset are highlighted in boldface in the first line, and the second line is the running time. It is seen clearly that the AUC values of the proposed LS^{3}T-Net exceed those of the seven benchmark methods. Considering dataset I as an example, the AUC (P_f, P_d) value of LS^{3}T-Net is 0.9940, which is 0.0537, 0.0276, 0.0230, 0.0629, 0.0498, 0.0124, and 0.0094 larger than those of RX, CRD, LSMAD, GTVLRR, PTA SparseHAD, and E2E-LIADE, respectively. Considering dataset II, the AUC (τ, P_f) value of LS^{3}T-Net is 0.0092, which is 0.0424, 0.0486, 0.0171, 0.1304, 0.1452, 0.0078, and 0.0281 smaller than those of RX, CRD, LSMAD, GTVLRR, PTA, SparseHAD, and E2E-LIADE, respectively. In addition, in Table III, it is clearly found that the deep learning-based methods need longer running time than the traditional methods. In this article, the numbers of training epochs of SparseHAD, E2E-LIADE, and LS^{3}T-Net are 3000, 100, and 100, respectively. E2E-LIADE and LS^{3}T-Net are more time-consuming because of the graph embedding and convolution operations. According to [46], Table IV lists the multiversion AUC values, in which lines 1–6 are AUC (τ, P_d), AUC_{BS}, AUC_{TD}, AUC_{OD}, AUC_{ODP}, and AUC_{SNPR} values, respectively. It is seen that the proposed LS^{3}T-Net can produce more stable performance for the six datasets.

C. Initial Detection Results of LS^{3}T-Net

Fig. 6 shows the corresponding initial and final detection maps for the six datasets. We found that $\mathbf{I}_{\text{up}}$ has a greater detection rate and $\mathbf{I}_{\text{down}}$ has smaller false alarm rate. For example, in dataset I, $\mathbf{I}_{\text{up}}$ could preserve the relative complete shape of the abnormal aircraft, and there are obvious brightness differences between the background and anomalies. In particular, the brightness of the abnormal aircraft is close to bright red and that of the background is dark blue. However, $\mathbf{I}_{\text{up}}$ has a greater false alarm rate, especially at the boundaries. The detection maps of spatial learning are usually affected by spatial context, and the boundary pixels are more likely to be regarded as anomalies because there are a larger spectral difference between the boundary pixels and the adjacent pixels. In Fig. 6(b1), the detected abnormal aircraft are incomplete, but the background is well suppressed. In Fig. 6(c1), $\mathbf{I}_{\text{final}}$ combines the advantages of spatial and spectral detection maps. It could not only preserve the abnormal shape well but also effectively suppress the false alarm rate. Fig. 7 shows the ROC curves based on the three detection maps, and Table V presents the corresponding AUC values. The results shown in Fig. 7 and Table V are consistent with the analysis in Fig. 6, and $\mathbf{I}_{\text{final}}$ shows a greater detection accuracy and a smaller false alarm rate.

In (13), the tradeoff parameter k can control the correction ability of $\mathbf{I}_{\text{down}}$. Fig. 8 shows the nonlinear function with different k values. We found that the larger the k values, the greater is the contribution of $\mathbf{I}_{\text{up}}$, while the smaller the k values, the closer $\mathbf{I}_{\text{final}}$ is to $\mathbf{I}_{\text{down}}$. Fig. 9 shows the final detection maps by different k values for dataset I. As shown in Fig. 9, with the increase of k value, the shape of the abnormal aircraft in the detection maps is more complete, and the brightness difference between the background and anomalies is greater. Meanwhile, the false alarm rate is also increasing. For example, the boundaries of the background are almost invisible in Fig. 9(a), but those are clear in Fig. 9(k). Therefore, to trade off preserving the abnormal shapes and

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TABLE IV

MULTIVERSION AUC VALUES OF THE EIGHT METHODS FOR THE SIX DATASETS (LINES 1–6 ARE AUC (τ, P_d), AUC_{BS}, AUC_{TD}, AUC_{OD}, AUC_{ODP}, AND AUC_{SNPR} VALUES, RESPECTIVELY)

	RX	CRD	LMSAD	GTVLRR	PTA	SparseHAD	E2E-LIADE	LS^{3}T-Net
Dataset I	0.1768	0.0723	0.2163	0.3447	0.4915	0.0172	0.1757	**0.4986**
	0.8815	0.9397	0.9550	0.8530	0.7632	0.9762	0.9722	**0.9823**
	1.1171	1.0387	1.1873	1.2758	1.4357	0.9988	1.1603	**1.4926**
	1.0583	1.0120	1.1713	1.1977	1.2547	0.9934	1.1479	**1.4809**
	1.1180	1.0456	1.2003	1.2666	1.3105	1.0118	1.1633	**1.4869**
	3.0068	2.7079	0.2003	4.4135	2.7154	3.1851	14.1693	**42.6153**
Dataset II	0.1731	0.2598	0.1512	0.3301	**0.3301**	0.1735	0.0807	0.3169
	0.7888	0.9180	0.9298	0.7305	0.7584	0.9402	0.8330	**0.9812**
	1.0135	1.2356	1.1073	1.2002	1.2429	1.1307	0.9510	**1.3073**
	0.9619	1.1778	1.0810	1.0606	1.0885	1.1137	0.9137	**1.2981**
	1.1215	1.2020	1.1249	1.1905	1.1757	1.1565	1.0434	**1.3077**
	3.3546	4.4948	5.7490	2.3646	2.1379	10.2059	2.1635	**34.4457**
Dataset III	0.0725	0.1713	0.3470	0.3998	**0.5909**	0.1322	0.3212	0.3297
	0.9280	0.9268	0.9406	0.8987	0.7583	0.9606	0.8216	**0.9850**
	1.0251	1.1480	1.3369	1.3863	**1.5626**	1.1068	1.2996	1.3246
	1.0005	1.0981	1.2876	1.2985	**1.3492**	1.0928	1.1428	1.3147
	1.0479	1.1214	1.2977	1.3120	**1.3775**	1.1182	1.1644	1.3198
	2.9472	3.4329	7.0385	4.5535	2.7690	9.4429	2.0485	**33.3030**
Dataset IV	0.3005	0.2865	0.2667	**0.5464**	0.5109	0.0431	0.3156	0.4842
	0.9353	0.9385	0.9375	0.7958	0.8459	**0.9868**	0.9091	0.9823
	1.2912	1.2812	1.2360	**1.5001**	1.4915	1.0354	1.2942	1.4808
	1.2358	1.2250	1.2042	1.3422	1.3568	1.0299	1.2247	**1.4665**
	1.2451	1.2303	1.2349	1.3885	1.3762	1.0376	1.2461	1.4699
	5.4242	5.0979	8.3868	3.4604	3.7929	7.8364	4.5410	33.8601
Dataset V	0.0686	0.2345	0.0724	0.2479	**0.5298**	0.0247	0.2239	0.3376
	0.8506	0.8931	**0.9818**	0.9315	0.7570	0.9661	0.9535	0.9500
	0.9571	1.2210	1.0629	0.2964	0.7602	0.9983	1.2149	**1.3325**
	0.9192	1.1276	1.0542	1.1794	1.2868	0.9908	1.1774	**1.2876**
	1.0307	1.1411	1.0637	1.1994	**1.2994**	1.0172	1.1864	1.2927
	1.8100	2.5107	**8.3218**	5.1113	2.2995	3.2933	5.9707	7.5189
Dataset VI	0.3507	0.3131	0.3425	0.3890	**0.4798**	0.0545	0.1236	0.4365
	0.9336	0.9460	0.9635	0.9360	0.6293	**0.9813**	0.6995	0.9787
	1.3346	1.3068	1.3327	1.3768	1.3230	1.0415	0.9124	**1.4269**
	1.2843	1.2591	1.3060	1.3250	1.1091	1.0358	0.8231	**1.4152**
	1.3004	1.2654	1.3158	1.3372	1.2659	1.0488	1.0343	**1.4248**
	6.9722	6.5639	12.8277	7.5097	2.2431	9.5614	1.3841	**37.3077**

suppressing the false alarm rate, the k value should be chosen as a compromise value.

D. Hyperparameter Analysis

Three main hyperparameters need to be tuned in the proposed LS^{3}T-Net: the patch size P and the number of hidden nodes of the upstream and downstream networks, K_1 and K_2. Fig. 10 shows the AUC values with the change of P and K_1 for the six datasets. It is observed that smaller P and larger K_1 values could produce larger AUC values. For example, for dataset I in Fig. 10(a), with the increase of P and the decrease of K_1, the AUC values clearly decrease. The remaining datasets show similar trends. Moreover, P significantly affects the AUC value. When P is 5 and K_1 was set to values in the candidate pool {128, 256, 512, 1024, 2048}, the corresponding AUC values were all around 0.98. However, when P increases to 11, the AUC values are around 0.95.

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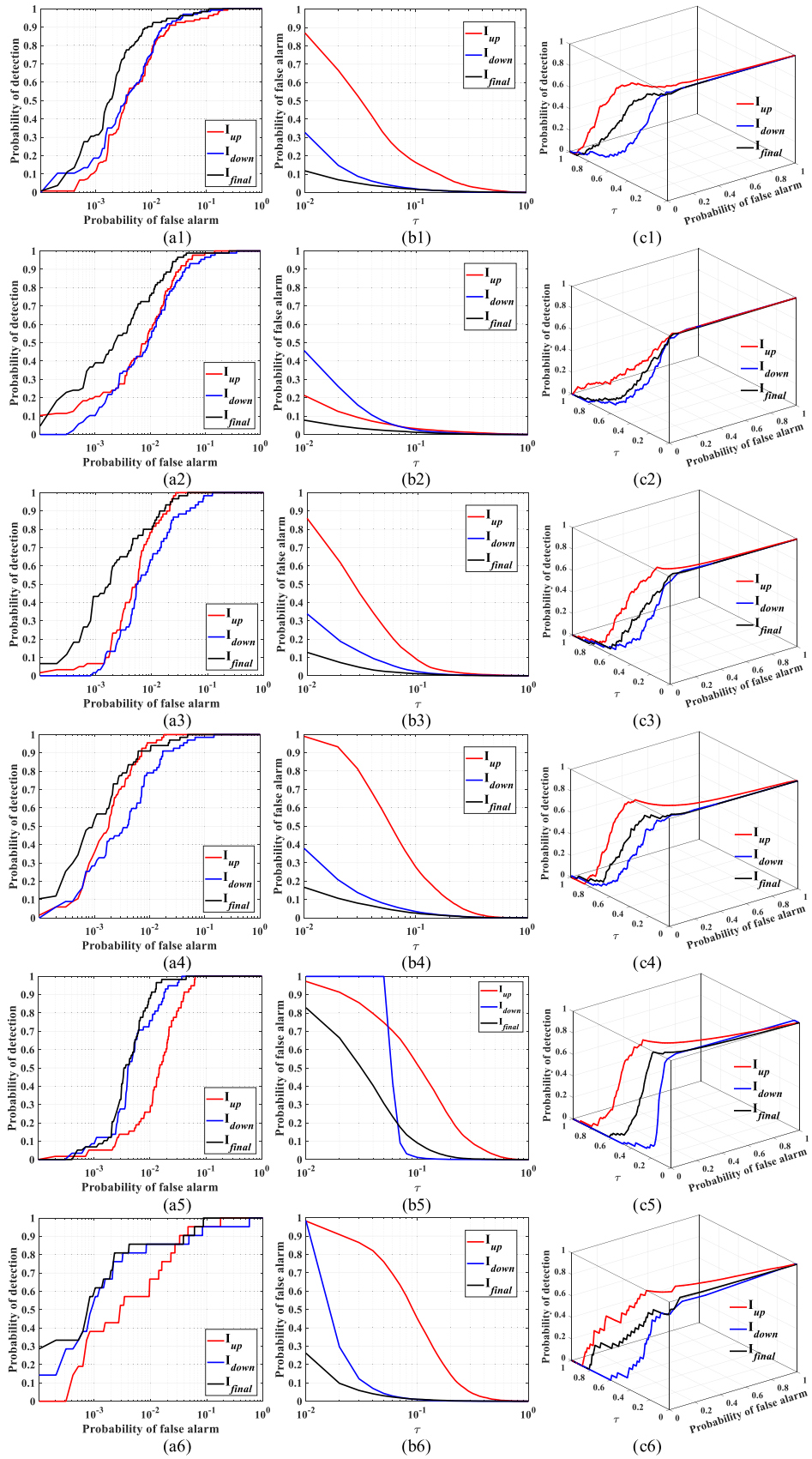


Fig. 7. ROC curves of different detection maps. (a) 2-D ROC curves of (P_f, P_d), (b) 2-D ROC curves of (τ, P_f), and (c) 3-D ROC curves. Lines 1–6 are the results for datasets I–VI, respectively.

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TABLE V
AUC VALUES $(P_d, P_f)/(P_f, \tau)$ OF THE INITIAL AND FINAL DETECTION RESULTS FOR THE SIX DATASETS

	$\mathbf{I}_{up}$	$\mathbf{I}_{down}$	$\mathbf{I}_{final}$
Dataset I	0.9854/0.0656	0.9906/0.0152	**0.9940/0.0117**
Dataset II	0.9853/0.0189	0.9784/0.0202	**0.9904/0.0092**
Dataset III	0.9930/0.0445	0.9843/0.0176	**0.9949/0.0099**
Dataset IV	**0.9971**/0.0892	0.9904/0.0198	0.9966/**0.0143**
Dataset V	0.9806/0.1396	0.9925/0.0612	**0.9949/0.0449**
Dataset VI	0.9824/0.1166	0.9652/0.0215	**0.9904/0.0117**

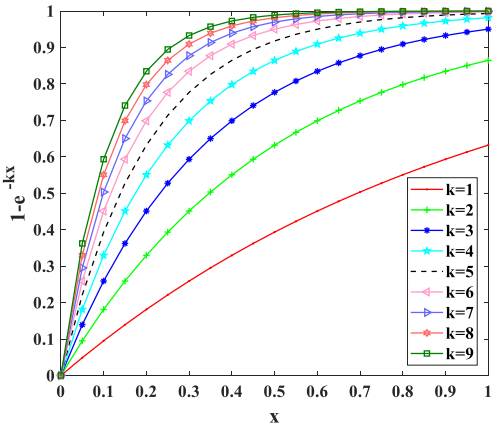


Fig. 8. Plot of the exponential constrained nonlinear functions with different k values.

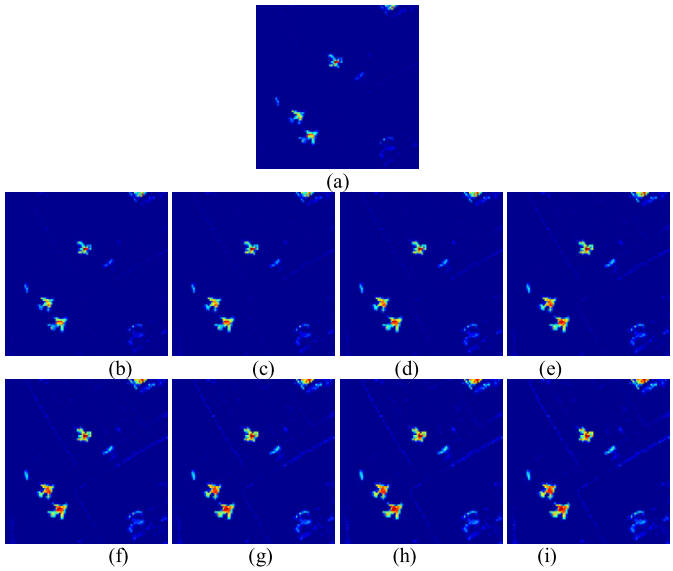


Fig. 9. Final maps of dataset I under different k values. (a)–(i) Detection maps with k varying from 1 to 9.

As a parameter of the downstream network, the number of hidden nodes K_2 was vital in the experiment results. Fig. 11 shows the AUC values under different K_2 values for the six datasets. We found that, except for dataset IV, the AUC values of most datasets are less affected by the K_2 value. In particular, the fluctuation of the AUC values for the six datasets is 0.0155, 0.0162, 0.0360, 0.0104, 0.0801, and 0.2098. In addition, for all the datasets, smaller K_2 values could produce larger AUC values.

E. Analysis of the Proportions of Training Samples

Dataset V was used as an example in this section. As shown in Fig. 12, Fig. 12(a) and (b) shows the false-color images (bands 37, 18, and 8 as RGB) and ground-reference map of anomalies, and Fig. 12(c)–(g) shows the binary maps with different proportions of the background samples. In these binary maps, the value of the candidate background pixel is 0, and the remaining pixel value is 1. It can be found that when the proportion is first 95%, most of the anomalies can be removed. With the decrease of the proportion, the anomalies can be generally removed, but some important background information will also be lost.

Fig. 13 shows the AUC values of the six datasets with different proportions of training samples. It can be seen that the detection accuracy increases with the increase of training samples. The main reason is that when the training samples are sufficient, the background distribution could be simply represented and facilitates the separation of background from anomalies. However, the excessively sufficient training samples might contain anomalies, which will affect modeling the background distribution.

V. DISCUSSION

Due to the lack of abnormal prior information, the challenge of hyperspectral anomaly detection as an unsupervised binary classification method is more difficult than the supervised version, namely, hyperspectral target detection. Most of the existing anomaly detection methods are semisupervised, and some potential background samples need to be coarsely selected in advance. In this article, we use the Mahalanobis distance to select potential background samples although it is not necessary to manually set parameters such as unsupervised clustering methods (such as K-means and DBSCAN). However, due to the influence of spectral variation, some anomalies may be misjudged as background pixels and mixed into the training samples. The impure background samples will affect the ability to model the background distribution. In future studies, we will try to construct more pure background samples by correcting spectral variation and further improve the learning ability of the semisupervised network.

In this article, the Mahalanobis distance was used as the final detection means. Although the Mahalanobis distance has been widely used in anomaly detection, as proposed in [4], [18], and [19], it is limited by the assumption that the background obeys the Gaussian distribution. When there are many anomalies or the background distribution is complex in HSIs, this method will be negatively affected. In addition, some deep learning-based methods use reconstruction error as the detection means [21]. This method is a challenge to the ability to extract background features. Therefore, it is important to design an effective detection method, which could fully separate the background and anomalies and is applicable to most HSIs. As VAE uses a probability encoder to model the distribution of hidden variables rather than variables themselves, the variability of the hidden space can be considered from the sampling process. It could be an interesting method based on combining reconstruction error with variability of hidden space.

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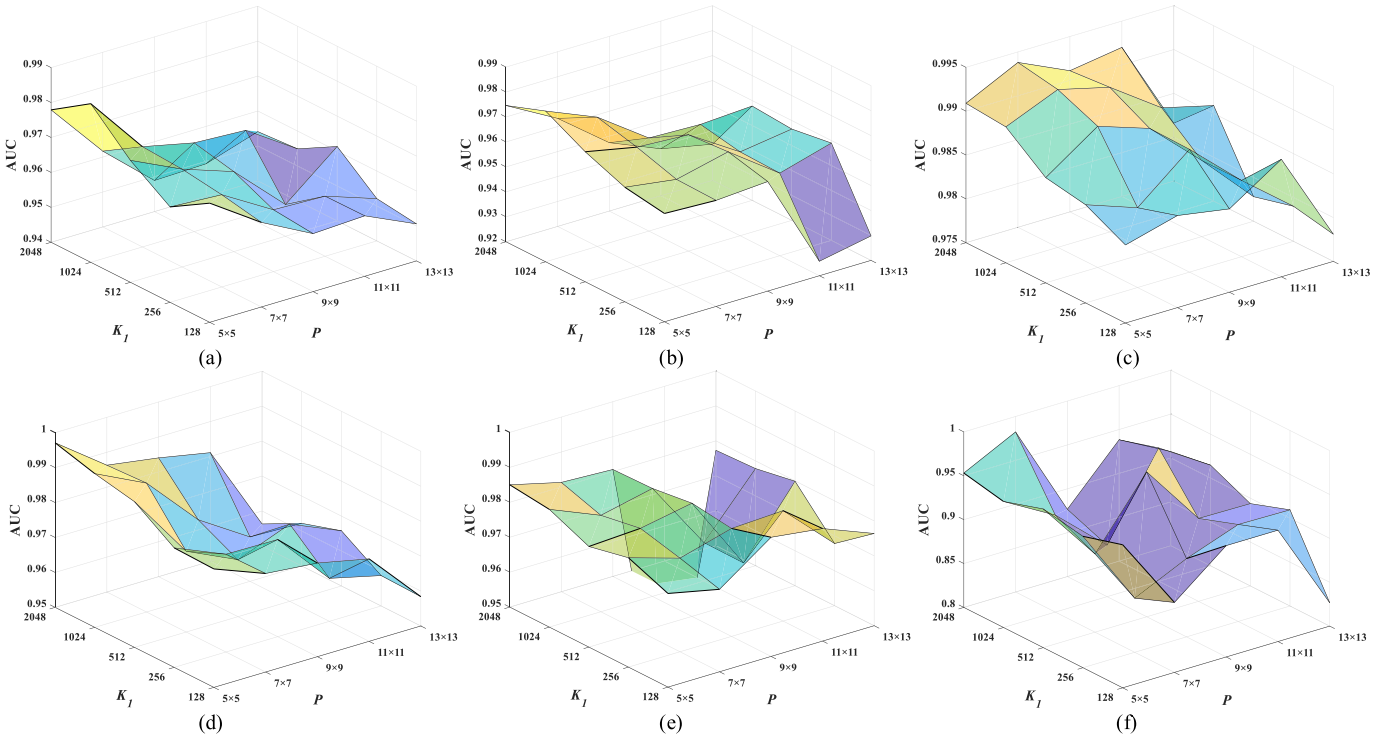


Fig. 10. AUC values of the upstream network with different values of P and K_1. (a) Dataset I. (b) Dataset II. (c) Dataset III. (d) Dataset IV. (e) Dataset V. (f) Dataset VI.

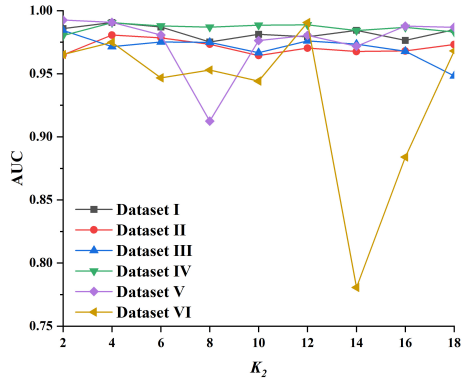


Fig. 11. AUC values of the downstream network with different K_2 values.

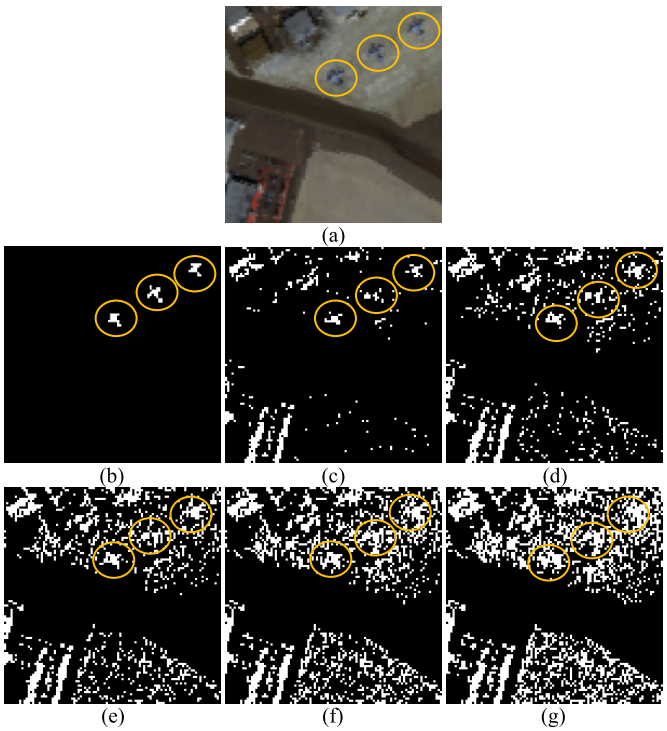


Fig. 12. Binary maps with different proportions of the background samples in dataset V. (a) False-color image (bands 37, 18, and 8 as RGB). (b) Ground-reference map of anomalies. (c)–(g) Binary maps with proportions of first 95%, 90%, 85%, 80%, and 75%, respectively.

In the upstream network of LS^{3}T-Net, image patch size is an important parameter. In the experiments, we tested the image patch size in the range of 5–13 pixels. It is found that a smaller image patch size could produce more reliability testing results. The main reason is that the smaller patch size has larger spatial autocorrelation, while the larger patch size might cover multiclass with great heterogeneity, even if all the pixels are background class. In addition, because HSIs have higher dimensions and smaller scales, overfitting is a common problem in deep learning-based methods. Molinier and Kilpi [47] considered that the training samples and test samples are not completely independent in HSI processing tasks. When using image patches as inputs, different samples show similarity due to the similar spatial context information, especially for larger patch size. We could select some image patches with strong spatial autocorrelation as training samples. However, this will lead to the problem of insufficient training samples. Data augmentation is one of the potential methods to suppress overfitting. There are some existing data augmentation methods based on image patch, such as mirroring, rotate, and add noise. Without destroying the spectral characteristics,

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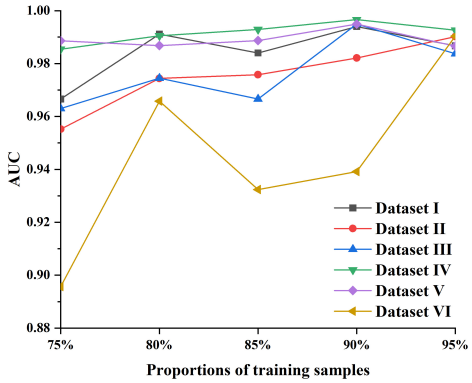


Fig. 13. AUC values with different proportions of training samples.

how to use the above methods to construct sufficient training samples is an urgent problem to be solved.

In addition to the nonlinear exponential constraint function, there are some other methods that have been applied to fuse spatial and spectral detection maps. For example, in [29], a linear function was used to fuse initial detection maps. Compared with nonlinear function, the linear function has acceptable interpretability. However, due to the complexity of data distribution, the results of the linear function are often not as competitive as those of nonlinear function. Moreover, both linear and nonlinear functions face the problem of setting parameters manually. In future work, it would be worthwhile to explore effective fusion functions and adaptive parameter setting methods.

The hidden space of VAE obeys the Gaussian distribution, and GMM has been used in hyperspectral anomaly detection [22], [48]. Using VAE as feature extraction network might be more conducive to the subsequent GMM to achieve the probability density estimation. In addition, in LS^{3}T-Net, the fusion process of spatial and spectral detection maps is operated outside the network. In future work, we will consider fusing spatial and spectral features within the network, such as applying additive, multiplicative, or concatenation fusion to realize feature fusion.

VI. Conclusion

In this article, we proposed an LS^{3}T-Net for hyperspectral anomaly detection. LS^{3}T-Net learns the background distribution from the perspective of local spatial and spectral information. LS^{3}T-Net not only effectively preserves the shape of abnormal targets but also improves the ability to separation between background and anomalies and reduces the false alarm rate. The results, including detection maps, box-whisker plots of anomalies and background, PR curves, and ROC curves coupled with AUC values, show that the proposed LS^{3}T-Net method is a satisfactory solution for hyperspectral anomaly detection, which is more accurate than seven benchmark methods.

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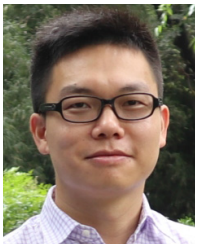
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